CampusOne Backend — Task & Verification Report

READY FOR HANDOFF

Phase 1 & Phase 2 Delivery — VESIT Multi-Tenant SuperApp Backend

1. Team Member Status Report

Name:	**Tanvi**
Assigned Task:	Task 1 — Dummy Student Database + Attendance Data (3,200 profiles, 16,000 attendance records, reproducible seeds)
Completed:	Generated 3,200 synthetic student profiles (exactly 800/year across FE, SE, TE, BE), 12 divisions/year across 6 branches, 60–70 students/division with FE smaller and SE larger distributions. Generated 16,000 consistent attendance records with validated constraints (`attended ≤ total`) and 2-decimal percentage formulas. Created migration 002, seeder utility, comprehensive coordination guide for Saumya, and 15-point automated verification suite.
Branch:	`feature/tanvi-dummy-students-attendance`
PR Link:	https://github.com/Sukrut-Mayee/vesit-superapp-backend/pull/new/feature/tanvi-dummy-students-attendance
Recent Commits:	`f6ceb54` docs: add student database and attendance seed specification and coordination guide `ec78fc5` test(seed): add 15-point verification suite for student counts, allocations, and attendance calculations `fdd015c` feat(schema): add migration 002 for student profile attributes and students table `19396d5` feat(seed): add deterministic generator and 3,200 student + 16,000 attendance demo datasets
Tests Passed:	**21/21 Unit Tests Passed (`npm test`) & 9/9 System Test Groups Passed (`node test_suite.js`)**
What is still failing:	**None.** 100% of unit tests, integrity checks, and system integration suites passed with 0 errors.
Blocked By:	**None.** All deliverables completed, tested, and pushed to remote branch.
Can next person start?	**YES** (Ready for Saumya and downstream teammates)

2. Requirements & Verification Checklist

- ✓ **3,200 Dummy Student Profiles:** Exactly 800 students per year across 4 academic years (FE, SE, TE, BE).
- ✓ **Branch & Division Allocations:** Computer (3), IT (3), EXTC (2), ECS (1), ARO (1), AIDS (2) = 12 divisions per year, 48 total.
- ✓ **Division Sizes (60–70):** Every single division strictly allocated between 60 and 70 students.
- ✓ **FE Smaller vs SE Larger Skew:** First-Year divisions start down at 61 students with 6 divisions ≤ 65; Second-Year divisions pushed up with 8 divisions ≥ 67 and all Computer/IT divisions at 68, preserving exact 800 totals.
- ✓ **100% Unique Identifiers:** 3,200 unique UUIDs, 3,200 unique roll numbers, 3,200 unique emails on `@vesit.ves.ac.in` (0 collisions).
- ✓ **16,000 Attendance Records:** 5 core subjects per student across branch curriculum; `attended ≤ total` guaranteed; exact 2-decimal percentage formula.
- ✓ **Repeatable Seed Generator:** Mulberry32 seeded PRNG produces byte-for-byte identical datasets on every run.
- ✓ **Safe Non-Destructive Migrations:** Added migration 002 using `ADD COLUMN IF NOT EXISTS` and `ON CONFLICT` idempotent guards.

3. Division & Student Distribution Matrix (Verified Counts)

Branch	Division	First Year (FE)	Second Year (SE)	Third Year (TE)	Final Year (BE)	Total Students
Computer (CMPN)	CMPN-A	65	68	67	67	**267**
Computer (CMPN)	CMPN-B	65	68	67	67	**267**
Computer (CMPN)	CMPN-C	64	68	67	67	**266**
Information Tech (IT)	IT-A	64	68	67	67	**266**
Information Tech (IT)	IT-B	63	68	67	67	**265**
Information Tech (IT)	IT-C	61 *(min)*	68	67	66	**262**
Electronics & Telecom (EXTC)	EXTC-A	69	67	67	67	**270**
Electronics & Telecom (EXTC)	EXTC-B	69	67	67	67	**270**
Electronics & Comp Sci (ECS)	ECS-A	70	65	66	66	**267**
Automation & Robotics (ARO)	ARO-A	70	65	66	66	**267**
AI & Data Science (AIDS)	AIDS-A	70	64	66	67	**267**
AI & Data Science (AIDS)	AIDS-B	70	64	66	66	**266**
TOTAL STUDENTS PER YEAR		**800**	**800**	**800**	**800**	**3,200**

4. Deliverable Files & Schema Specification (Handoff for Saumya)

All seed artifacts are generated and persisted directly in the repository for immediate consumption by downstream modules:

File Path	Format	Size / Records	Description
`data/students_seed.json`	JSON	1.8 MB (3,200 items)	Full student objects: ID, name, email, roll number, year, branch, division, college_id, password_hash
`data/attendance_seed.json`	JSON	8.1 MB (16,000 items)	Full attendance records: student_id, roll_number, subject, attended, total, percentage, sem, year
`data/seed_students_and_attendance.sql`	SQL Script	4.0 MB (Batch insert)	Repeatable, idempotent SQL insert with `ON CONFLICT DO UPDATE`
`data/students_seed.csv`	CSV	744 KB (3,200 rows)	Tabular student data for spreadsheets or ingestion scripts
`data/attendance_seed.csv`	CSV	3.6 MB (16,000 rows)	Tabular attendance records
`docs/STUDENT_AND_ATTENDANCE_SEED.md`	Markdown	10 KB	Complete schema contract, subject syllabus mapping, and coordination guide

5. Schema Payloads & Attendance Formulas

Student Profile Schema (`users` / `students`):


```
{ "id": "53e74ec1-cfc3-4152-a0c9-35d08c5cd93e", "college_id": "81055c6a-4909-44e3-88a3-be6cff32635a", "name": "Aarav Sharma", "email": "aarav.sharma.fecmpna01@vesit.ves.ac.in", "password_hash": "$2a$10$etLP...", // Password123! "role": "student", "trust_score": 12, "roll_number": "FE-CMPN-A-01", "year": "First Year", "year_code": "FE", "branch": "Computer", "branch_code": "CMPN", "division": "CMPN-A", "division_letter": "A" }
```


Attendance Record Schema (`attendance`):


```
{ "id": "8cb69823-3765-42ec-99bc-3b10b0e50974", "college_id": "81055c6a-4909-44e3-88a3-be6cff32635a", "student_id": "53e74ec1-cfc3-4152-a0c9-35d08c5cd93e", "roll_number": "FE-CMPN-A-01", "subject": "Engineering Mathematics I", "attended_classes": 32, "total_classes": 36, "percentage": 88.89, "semester": "SEM 1", "division": "CMPN-A", "academic_year": "2026-2027" }
```


6. Execution & Verification Commands

To run, regenerate, or verify the seed datasets locally:

```
# 1. Regenerate 100% reproducible dummy data (Mulberry32 PRNG seed) node scripts/generate_demo_data.js # 2. Run complete automated 21-test verification suite npm test # 3. Dry-run database seeder without live DB queries node scripts/seed_database.js --dry-run # 4. Verify existing record counts in Supabase database node scripts/seed_database.js --verify-only
```